

Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement 7: Hostel Maintenance & Issue Ticketing System

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Section: 5H**1. Requirements Table**

Stakeholders / Actors: Hostel Resident, Maintenance Warden, Hostel Administrator.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall enable residents to log maintenance tickets with room number tags, photo attachments, and categorized issue severity.	High	Pass: Ticket is created, assigned a unique ID, and routed to warden queue. Fail: Incomplete form accepted without room tag.	Core intake function; ensures maintenance staff receive complete, actionable ticket information.
FR-002	Functional	The system shall allow the Maintenance Warden to assign a submitted ticket to an available maintenance staff member based on issue category (plumbing / electrical / carpentry).	High	Pass: Ticket status changes to "Assigned" and the assigned staff member's name/ID appears in the record within 5 minutes of submission. Fail: Ticket remains unassigned after 5 minutes with no staff notified.	Ensures timely dispatch and category-appropriate routing of maintenance work.
FR-003	Functional	The system shall allow residents to view the real-time status (Open, Assigned, In Progress, Resolved, Closed) of any ticket they have logged.	Medium	Pass: Status field updates and is visible to the resident within 1 minute of any warden action. Fail: Resident sees outdated or blank status.	Improves transparency and reduces repeat status-inquiry calls to the hostel office.
FR-004	Functional	The system shall allow the Maintenance Warden to update a ticket's status and add resolution notes with a timestamp at each stage of the repair process.	High	Pass: Each status change is logged with warden ID, timestamp, and notes, and is retrievable in the ticket history. Fail: Status change is saved without a timestamp or warden ID.	Provides an auditable trail required for SLA tracking and escalation.
FR-005	Functional	The system shall require the resident to provide a closure sign-off (confirmation and an optional satisfaction rating) before a ticket marked "Resolved" can move to "Closed".	Medium	Pass: Ticket status changes to "Closed" only after resident confirmation is recorded. Fail: Ticket auto-closes without resident confirmation within the SLA window.	Confirms genuine issue resolution from the resident's perspective and prevents premature closure.
NFR-001	Nonfunctional	The system shall trigger automated escalation SMS/Email alerts if high-priority tickets exceed a 24-hour SLA without a status change.	High	Pass: Benchmarking tests confirm target latency and security standards are met under simulated peak load.	Prevents high-priority issues from being neglected and keeps hostel administration accountable to SLA commitments.
NFR-002	Nonfunctional	The system shall encrypt all stored photo attachments and resident personal data at rest and in transit, and shall restrict ticket	High	Pass: A security audit / penetration test confirms attachments and personal data are encrypted and	Protects resident privacy and meets institutional data-protection

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		access to the filing resident, the assigned warden, and authorized administrators.		inaccessible to unauthorized roles. Fail: Any unauthorized role can view resident personal data or photos.	requirements, since tickets contain room numbers, photos, and contact details.

2. UML Use-Case Diagram



3. Use-Case Flow Specification

Use Case: UC-01 – Log Maintenance Ticket

Primary Actor: Hostel Resident

Preconditions: The resident is logged into the hostel maintenance app/portal with valid hostel credentials, and has an assigned room number on file in the system.

Postconditions (success): A new ticket exists with a unique Ticket ID, status “Open,” and is placed in the Maintenance Warden's queue for assignment.

Main Success Scenario:

- 1. Resident selects “Log New Ticket” from the app/portal home screen.
- 2. System displays a ticket form requesting room number (pre-filled/geo-tagged), issue category (plumbing / electrical / carpentry), and severity (Low / Medium / High).
- 3. Resident selects the issue category and severity level.
- 4. Resident optionally attaches one or more photos of the issue (invokes UC-02: Attach Photo Evidence).
- 5. Resident enters a short description of the problem and submits the form.
- 6. System validates that room number, category, and severity are all present.
- 7. System generates a unique Ticket ID and timestamps the submission.
- 8. System sets the ticket status to “Open” and routes it to the Maintenance Warden's queue.
- 9. System displays a confirmation message containing the Ticket ID to the resident.
- 10. Use case ends successfully.

Alternate Flow – 6a. Missing / Invalid Room Tag:

- 6a1. System detects that the room number field is empty or invalid.
- 6a2. System displays an error message: “Room number is required to submit a ticket.”
- 6a3. Resident is returned to the form to correct the room number.
- 6a4. Resident re-submits the form; flow resumes at step 6. (If the resident abandons the form instead, no ticket is created and the use case ends in failure.)